

## Module-2

### Session-1

Q1. Write a short paragraph in your own words defining what a 'program' is and give one real-life example from any app you use daily (like Instagram, Zomato, or Paytm).

Ans. A program is a set of instructions given to a computer or mobile phone to perform a specific task.

Programs help us use different apps and complete our daily work easily. For example, when we use **Zomato**.

a program allows us to search for restaurants, choose food, add items to the cart, place the order, make payment, and track the delivery.

Q2. List three popular programming languages you have heard of, and for each, mention whether it is usually compiled, interpreted, or can be both.<br><br><em><strong>Hint:</strong> Think of languages like Java, Python, and JavaScript.</em>

Ans. **Java – Usually Compiled**

Java is a popular programming language used to create mobile apps, software, and web applications. Java programs are usually **compiled** into bytecode before they are executed.

**Python – Usually Interpreted**

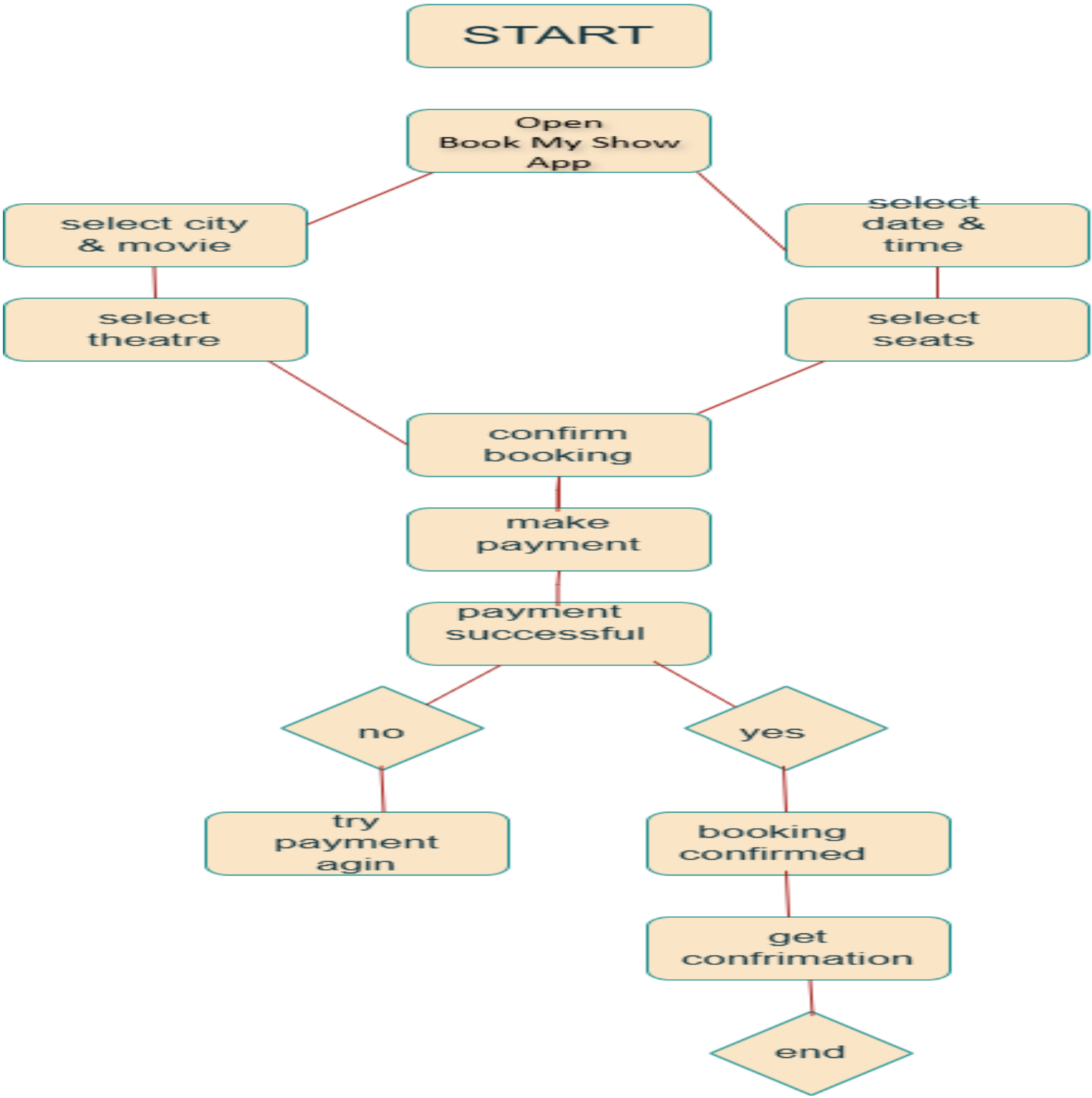
Python is a simple and popular programming language. It is used in websites, data science, artificial intelligence, and automation. Python is usually considered an **interpreted** language.

**JavaScript – Can Be Both**

JavaScript is mainly used to make websites interactive. It is also used in web applications and some mobile and desktop applications. Modern JavaScript engines can use both interpretation and compilation techniques.

Q3. Draw a flowchart (on paper or using any online tool) for the process of booking a movie ticket on Book My Show, starting from opening the app to receiving the booking confirmation.

Ans.



Q4. Write down the steps (as an algorithm, using simple numbered instructions) for ordering food on Swiggy — from opening the app to completing payment.

Ans. **Start**

Open the **Swiggy app**.

Select your **location**.

Search for a **restaurant or food item**.

Select the food you want.

Add the food to the **cart**.

Open the cart and **check your order**.

Enter or confirm the **delivery address**.

Select a **payment method**.

Make the **payment**.

Payment successful → **Order Confirmed**.

**End.**

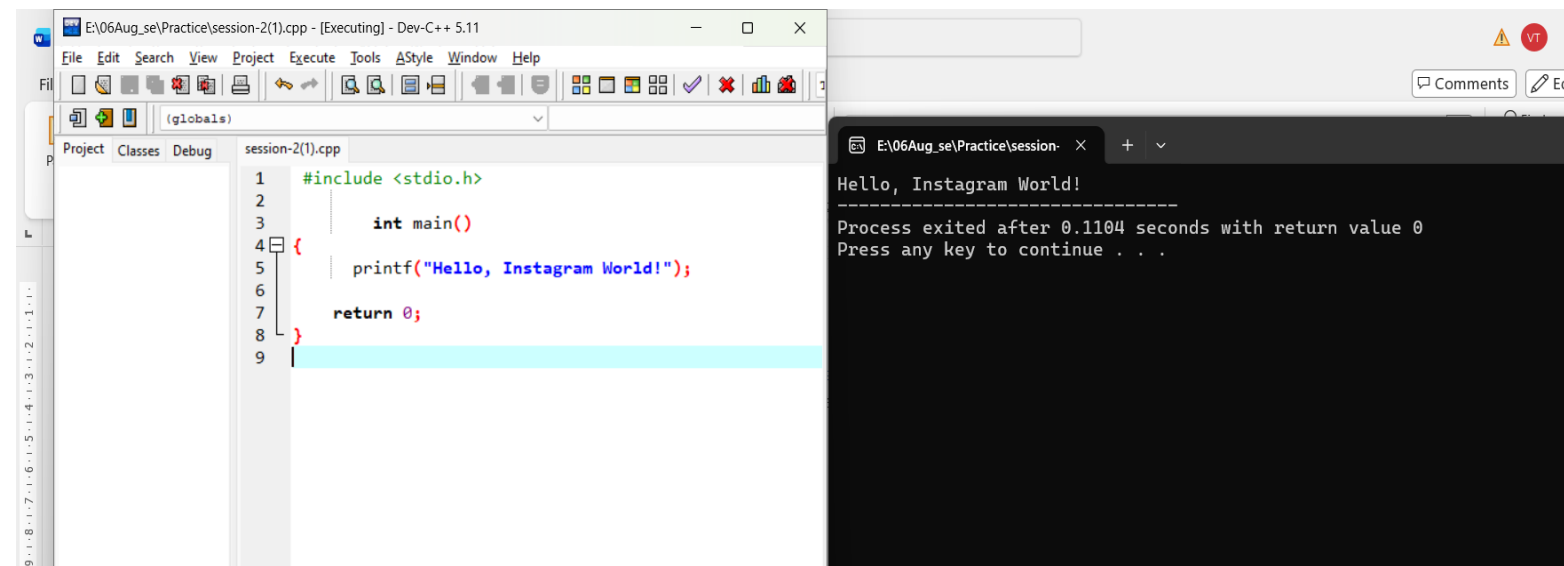
## Session-2

Q1. Install GCC on your computer and verify the installation by running `gcc --version` in your terminal or command prompt.

Ans. Install GCC on your computer. Open **Command Prompt/Terminal** and type:

```
gcc --version
```

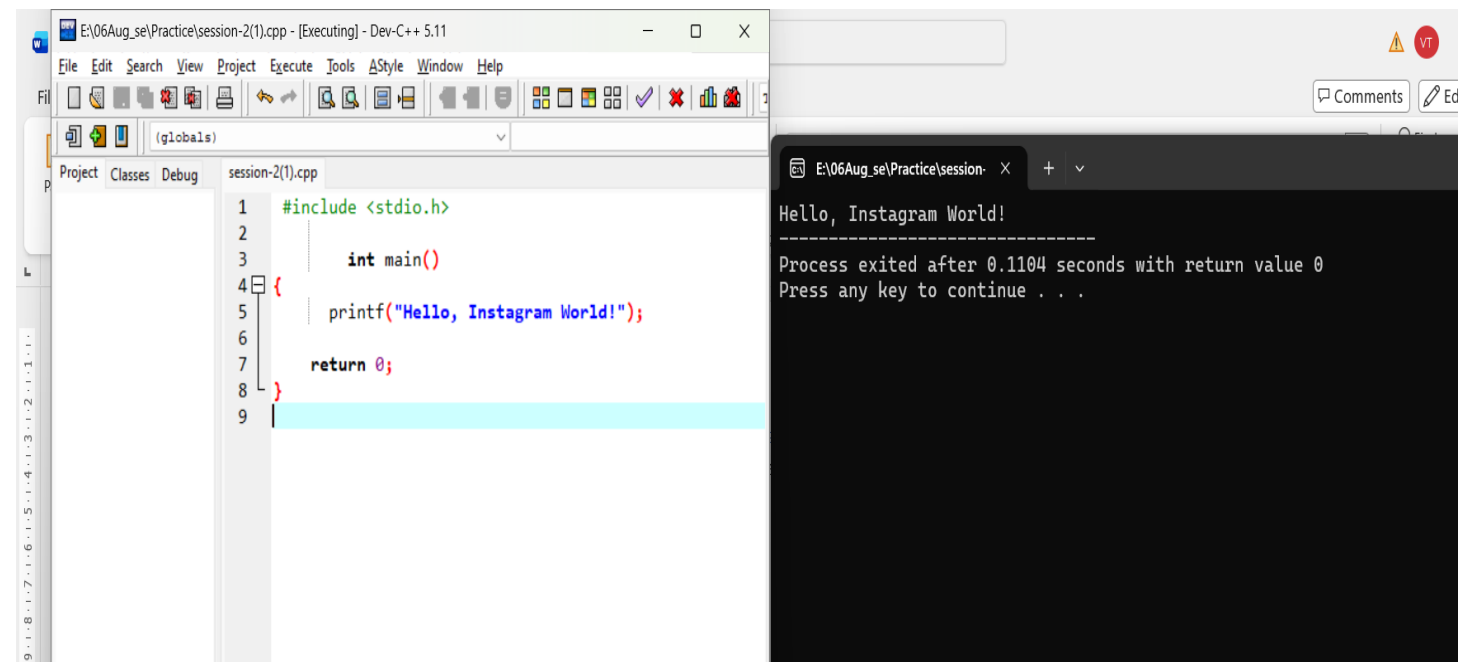
Q2. Open VS Code or Dev C++ and create a new file named `hello.c`. Write a C program that prints 'Hello, Instagram World!' to the console.



Hello, Instagram World!

Q3. Identify and label the header, main function, and return statement in your hello.c program using comments.

Ans



The screenshot displays the Dev-C++ IDE interface. The left pane shows the source code for 'session-2(1).cpp' with the following content:

```
1  #include <stdio.h>
2
3      int main()
4  {
5      printf("Hello, Instagram World!");
6
7      return 0;
8  }
9
```

The right pane shows the execution output in a terminal window. The output is as follows:

```
Hello, Instagram World!
-----
Process exited after 0.1104 seconds with return value 0
Press any key to continue . . .
```

**Header:** #include <stdio.h>

**Main Function:** int main()

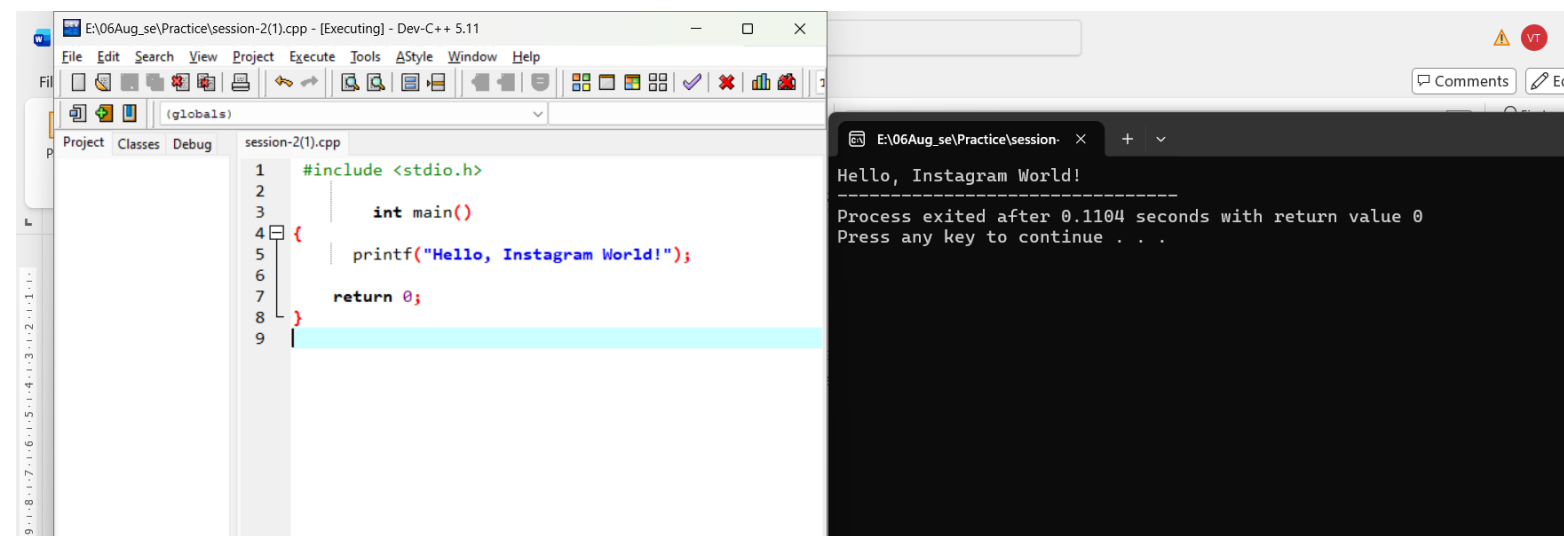
**Return Statement:** return 0;

Q4. Compile your hello.c file using the GCC command line (gcc hello.c -o hello) and run the output file to display the message in your terminal.<br><br><em><strong>Hint:</strong></em> On Windows, the output file will be hello.exe; on Linux/Mac, it will be hello.</em>

Ans.

**Answer:**

- **Header:** #include <stdio.h>
- **Main Function:** int main()
- **Return Statement:** return 0;



The screenshot displays the Dev-C++ IDE interface. The main editor window shows a C program named 'session-2(1).cpp' with the following code:

```
1 #include <stdio.h>
2
3     int main()
4 {
5     printf("Hello, Instagram World!");
6
7     return 0;
8 }
9
```

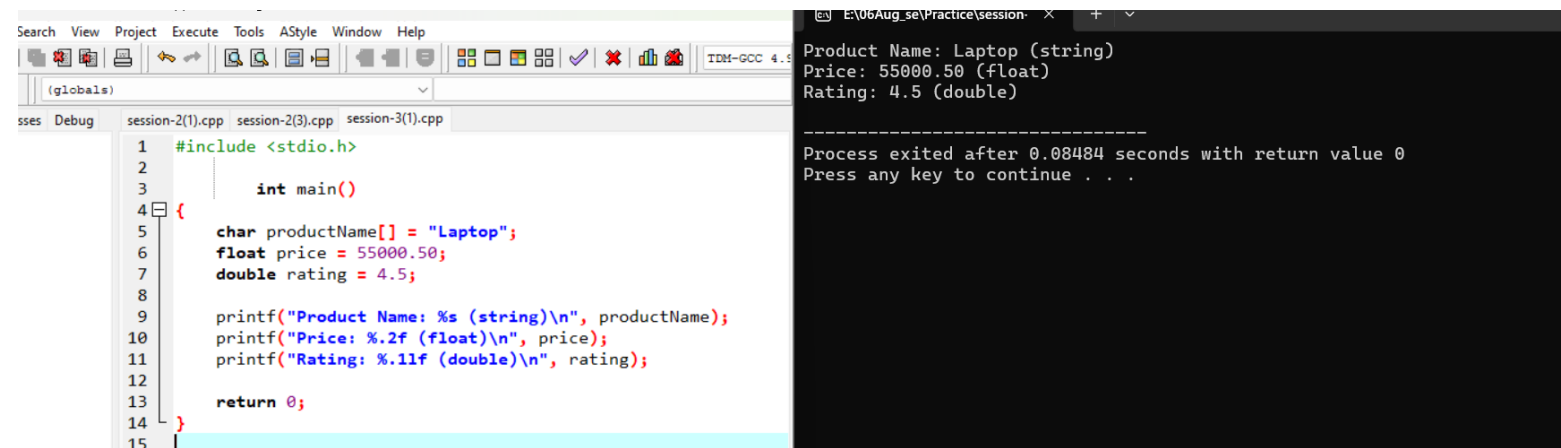
The code is syntactically highlighted. The execution output window on the right shows the program's runtime behavior:

```
Hello, Instagram World!
-----
Process exited after 0.1104 seconds with return value 0
Press any key to continue . . .
```

## Session-3

Q1. Declare variables for a Flipkart product: product Name (as a string), price (float), and rating (double). Assign sample values and print each variable with its data type.

Ans.



The screenshot displays a C++ IDE with two panels. The left panel shows the source code for a program that declares and prints variables for a Flipkart product. The right panel shows the output of the program, which matches the code's intent.

```
#include <stdio.h>

int main()
{
    char productName[] = "Laptop";
    float price = 55000.50;
    double rating = 4.5;

    printf("Product Name: %s (string)\n", productName);
    printf("Price: %.2f (float)\n", price);
    printf("Rating: %.11f (double)\n", rating);

    return 0;
}
```

```
Product Name: Laptop (string)
Price: 55000.50 (float)
Rating: 4.5 (double)

-----
Process exited after 0.08484 seconds with return value 0
Press any key to continue . . .
```

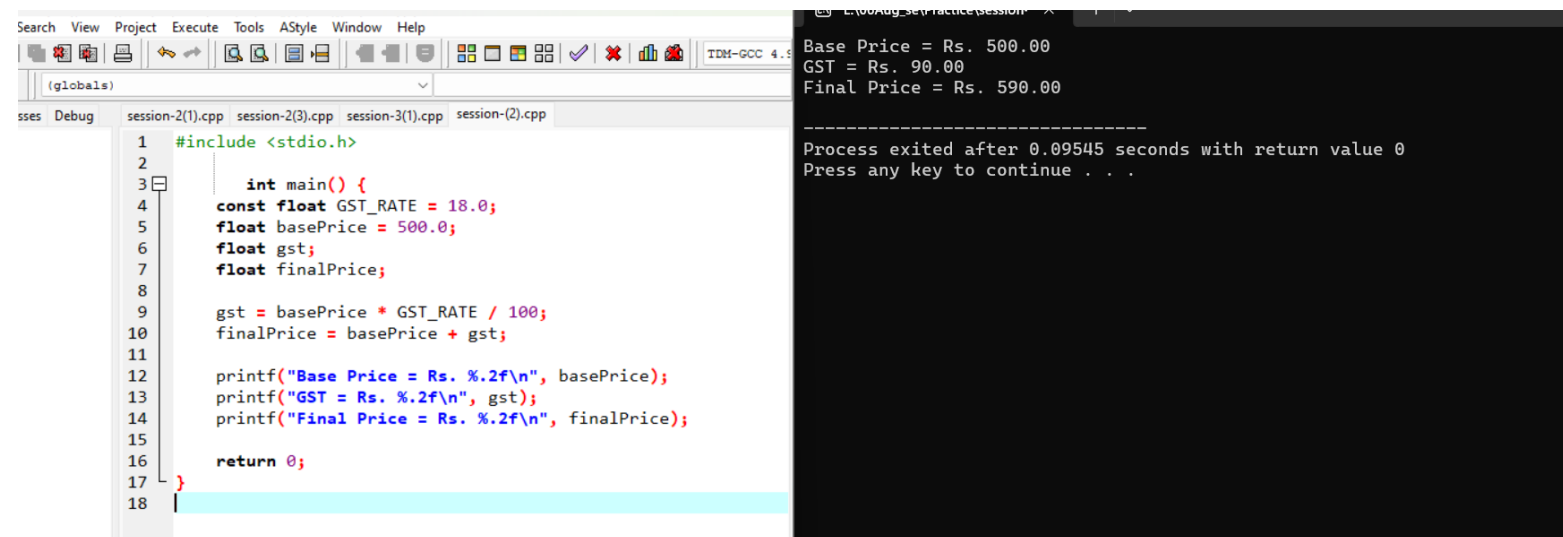
Product Name: Laptop (string)

Price: 55000.50 (float)

Rating: 4.5 (double)

Q2. Create a constant variable to store the GST rate (for example, 18%) and use it to calculate the final price of a Zomato order with a given base price.<br><br><em><strong>Constraint:</strong> The GST rate must not be changeable after its initial assignment.</em>

Ans



The screenshot shows a C++ IDE with a code editor on the left and a terminal on the right. The code in the editor defines a constant GST rate of 18% and calculates the final price of a Zomato order with a base price of 500.00. The terminal output shows the calculated values: Base Price = Rs. 500.00, GST = Rs. 90.00, and Final Price = Rs. 590.00.

```
#include <stdio.h>

int main() {
    const float GST_RATE = 18.0;
    float basePrice = 500.0;
    float gst;
    float finalPrice;

    gst = basePrice * GST_RATE / 100;
    finalPrice = basePrice + gst;

    printf("Base Price = Rs. %.2f\n", basePrice);
    printf("GST = Rs. %.2f\n", gst);
    printf("Final Price = Rs. %.2f\n", finalPrice);

    return 0;
}
```

```
Base Price = Rs. 500.00
GST = Rs. 90.00
Final Price = Rs. 590.00

-----
Process exited after 0.09545 seconds with return value 0
Press any key to continue . . .
```

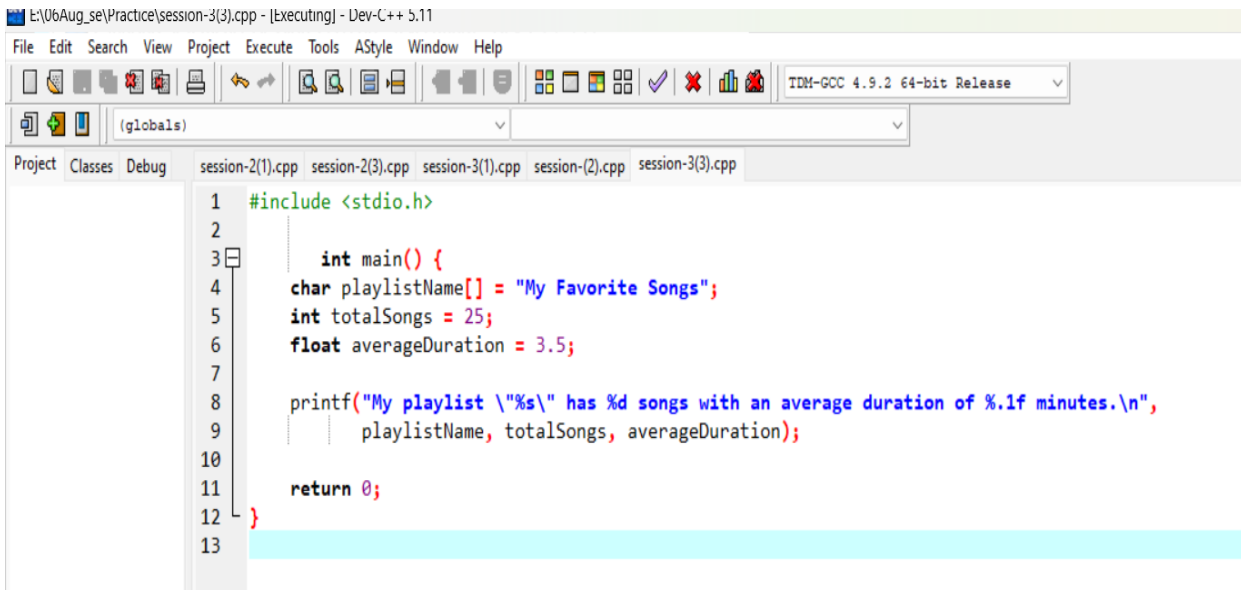
Base Price = ₹500

GST = ₹90

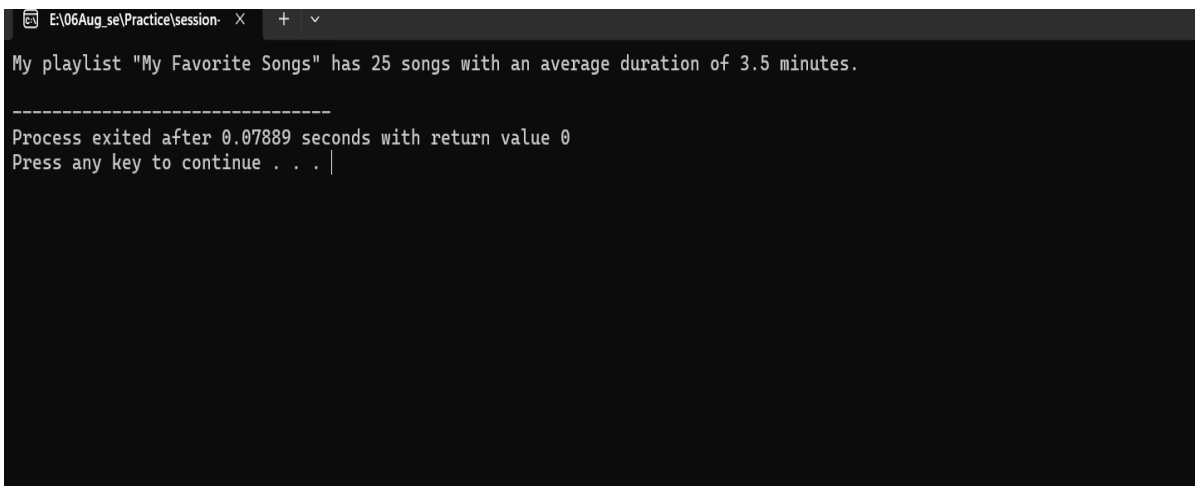
Final Price = ₹590

Q3. Write a program that stores your favorite Spotify playlist's name (string), total number of songs (int), and average song duration in minutes (float). Print all values in a single formatted sentence.

Ans.



```
1 #include <stdio.h>
2
3 int main() {
4     char playlistName[] = "My Favorite Songs";
5     int totalSongs = 25;
6     float averageDuration = 3.5;
7
8     printf("My playlist \"%s\" has %d songs with an average duration of %.1f minutes.\n",
9           playlistName, totalSongs, averageDuration);
10
11     return 0;
12 }
13
```



```
My playlist "My Favorite Songs" has 25 songs with an average duration of 3.5 minutes.
-----
Process exited after 0.07889 seconds with return value 0
Press any key to continue . . .
```

Playlist Name → My Favorite Songs

- Total Songs → 25
- Average Duration → 3.5 minutes

Q4. Identify and correct the invalid variable names from this list: 1stPlayer, player\_age, \$score, total-marks, userName.

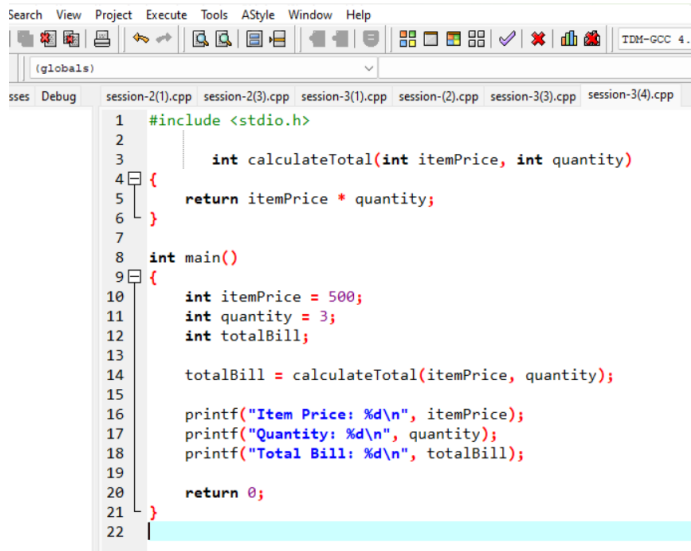
**Hint:** Apply standard variable naming rules for your programming language.

|                |   |             |
|----------------|---|-------------|
| Ans. 1stPlayer | → | firstPlayer |
| player_age     | → | player_age  |
| \$score        | → | score       |
| total-marks    | → | total_marks |
| userName       | → | userName    |

## Session-4

Q1. Create a simple JavaScript function called `calculateTotal` that takes two numbers: `itemPrice` and `quantity`, and returns the total bill amount using arithmetic operators.

Ans.



```
1 #include <stdio.h>
2
3     int calculateTotal(int itemPrice, int quantity)
4 {
5     return itemPrice * quantity;
6 }
7
8 int main()
9 {
10     int itemPrice = 500;
11     int quantity = 3;
12     int totalBill;
13
14     totalBill = calculateTotal(itemPrice, quantity);
15
16     printf("Item Price: %d\n", itemPrice);
17     printf("Quantity: %d\n", quantity);
18     printf("Total Bill: %d\n", totalBill);
19
20     return 0;
21 }
22
```

```
Item Price: 500
Quantity: 3
Total Bill: 1500
```

```
-----
Process exited after 0.09887 seconds with return value 0
Press any key to continue . . .
```

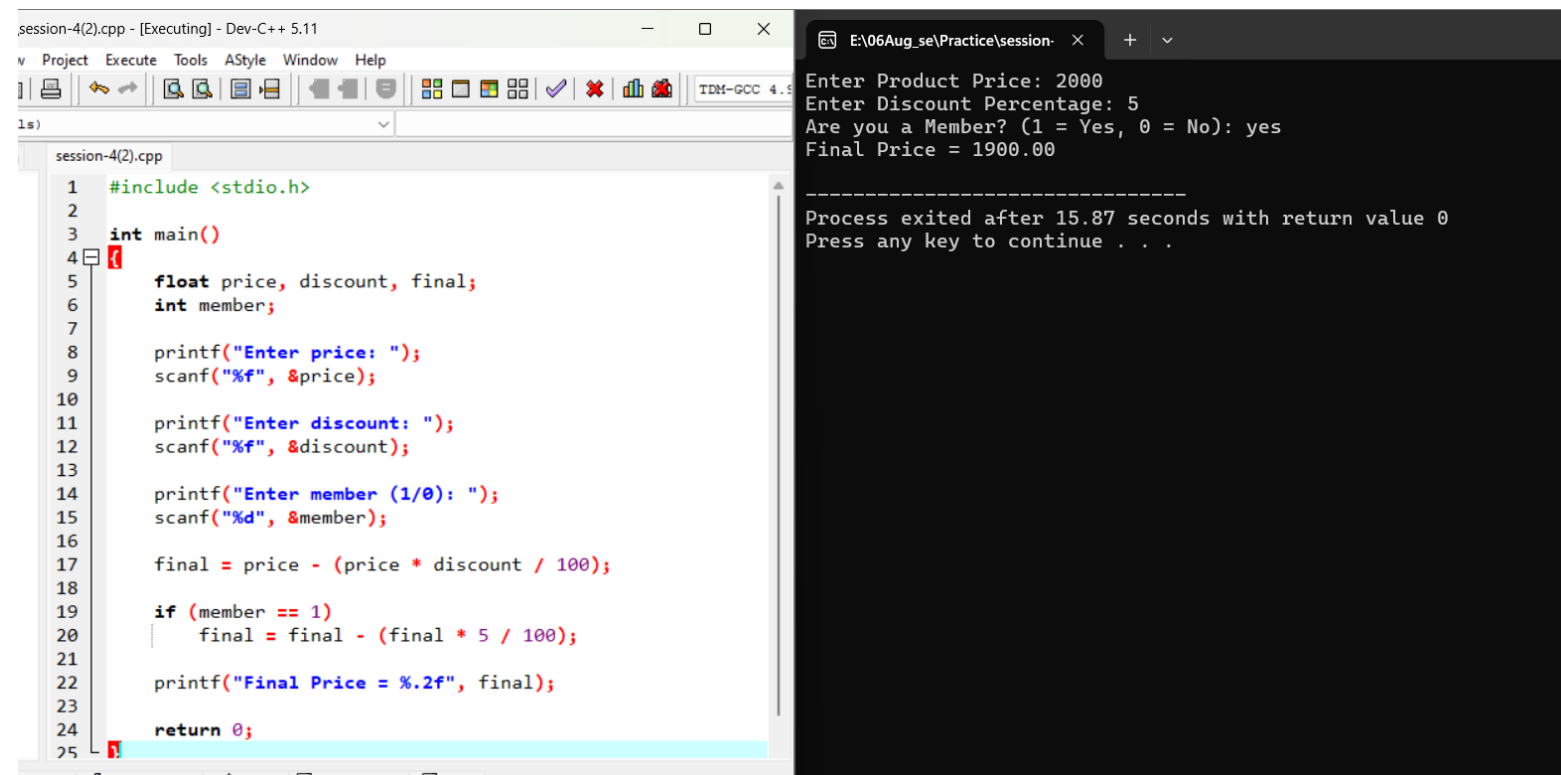
Item Price: 500

Quantity: 3

Total Bill: 1500

Q2. Build a Flipkart-style discount calculator: given product price, discount percentage, and a boolean is Member, use arithmetic and logical operators to calculate the final price (apply an extra 5% off if isMember is true).

Ans.



The image shows a screenshot of a C++ program being executed in Dev-C++. The program is named 'session-4(2).cpp' and is located in the directory 'E:\06Aug\_se\Practice\session-'. The code is as follows:

```
1 #include <stdio.h>
2
3 int main()
4 {
5     float price, discount, final;
6     int member;
7
8     printf("Enter price: ");
9     scanf("%f", &price);
10
11    printf("Enter discount: ");
12    scanf("%f", &discount);
13
14    printf("Enter member (1/0): ");
15    scanf("%d", &member);
16
17    final = price - (price * discount / 100);
18
19    if (member == 1)
20        final = final - (final * 5 / 100);
21
22    printf("Final Price = %.2f", final);
23
24    return 0;
25 }
```

The terminal output shows the program's execution with the following inputs and outputs:

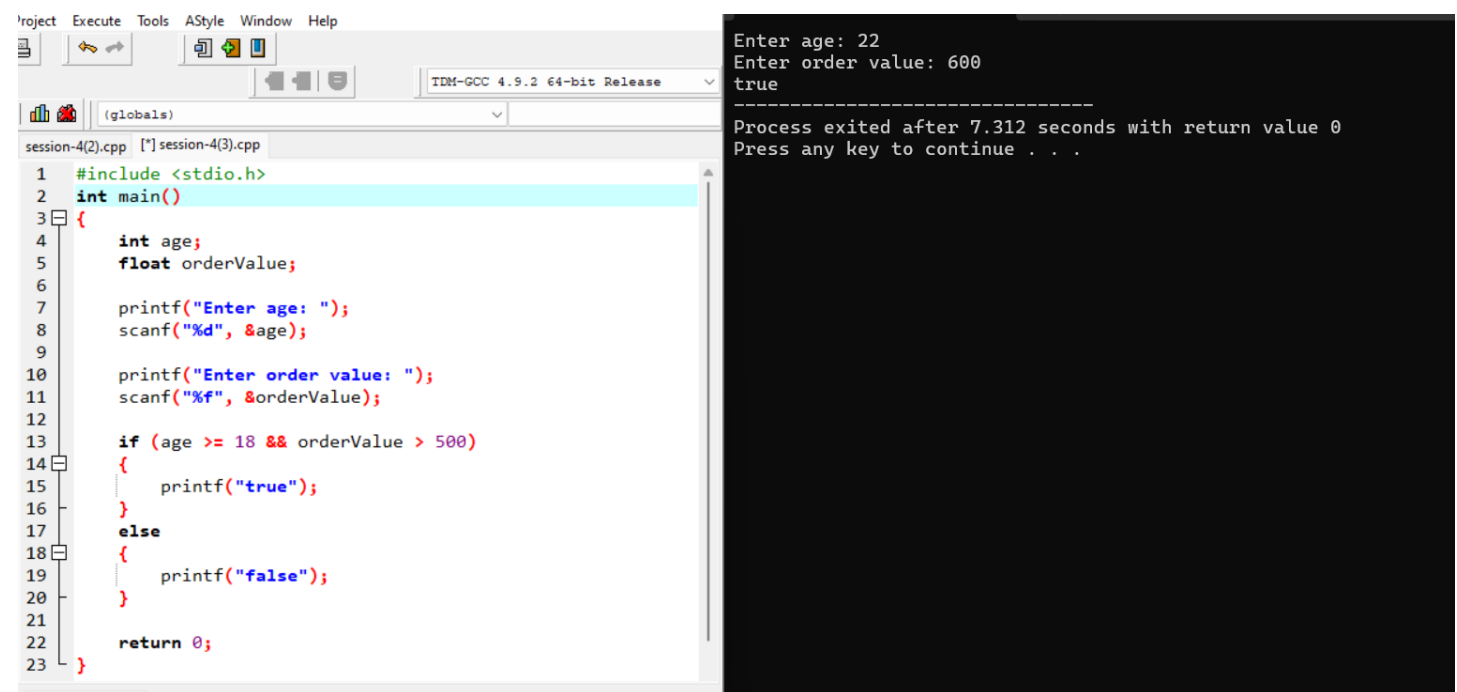
```
Enter Product Price: 2000
Enter Discount Percentage: 5
Are you a Member? (1 = Yes, 0 = No): yes
Final Price = 1900.00

-----
Process exited after 15.87 seconds with return value 0
Press any key to continue . . .
```

Enter Product Price: 2000  
Enter Discount Percentage: 5  
Are you a Member (1 = Yes, 0 = No): yes  
Final Price = 1900.00

Q3. Write a function isEligibleForOffer that takes a user's age and total order value, and returns true if the user is 18 or older AND the order value is above 500, otherwise false.<br><br><em><strong>Hint:</strong> Use relational and logical operators together.</em>

Ans.



The screenshot shows a C++ IDE with a source code editor on the left and a terminal window on the right. The source code in the editor is as follows:

```
1 #include <stdio.h>
2 int main()
3 {
4     int age;
5     float orderValue;
6
7     printf("Enter age: ");
8     scanf("%d", &age);
9
10    printf("Enter order value: ");
11    scanf("%f", &orderValue);
12
13    if (age >= 18 && orderValue > 500)
14    {
15        printf("true");
16    }
17    else
18    {
19        printf("false");
20    }
21
22    return 0;
23 }
```

The terminal window on the right shows the following output:

```
Enter age: 22
Enter order value: 600
true
-----
Process exited after 7.312 seconds with return value 0
Press any key to continue . . .
```

Enter age: 22

Enter order value: 600

True

Q4. Given three variables: likes, comments, and shares (all numbers), write code to check if a post is 'trending' on Instagram (at least 1000 likes OR more than 200 comments AND at least 50 shares). Print the result.

Ans. #include <stdio.h>

int main()

{

int likes, comments, shares;

printf("Enter likes: ");

```
scanf("%d", &likes);
```

```
printf("Enter comments: ");
```

```
scanf("%d", &comments);
```

```
printf("Enter shares: ");
```

```
scanf("%d", &shares);
```

```
if (likes >= 1000 || (comments > 200 && shares >= 50))
```

```
{
```

```
    printf("Trending");
```

```
}
```

```
else
```

```
{
```

```
    printf("Not Trending");
```

```
}
```

```
return 0;
```

```
}
```

Enter likes: 1200

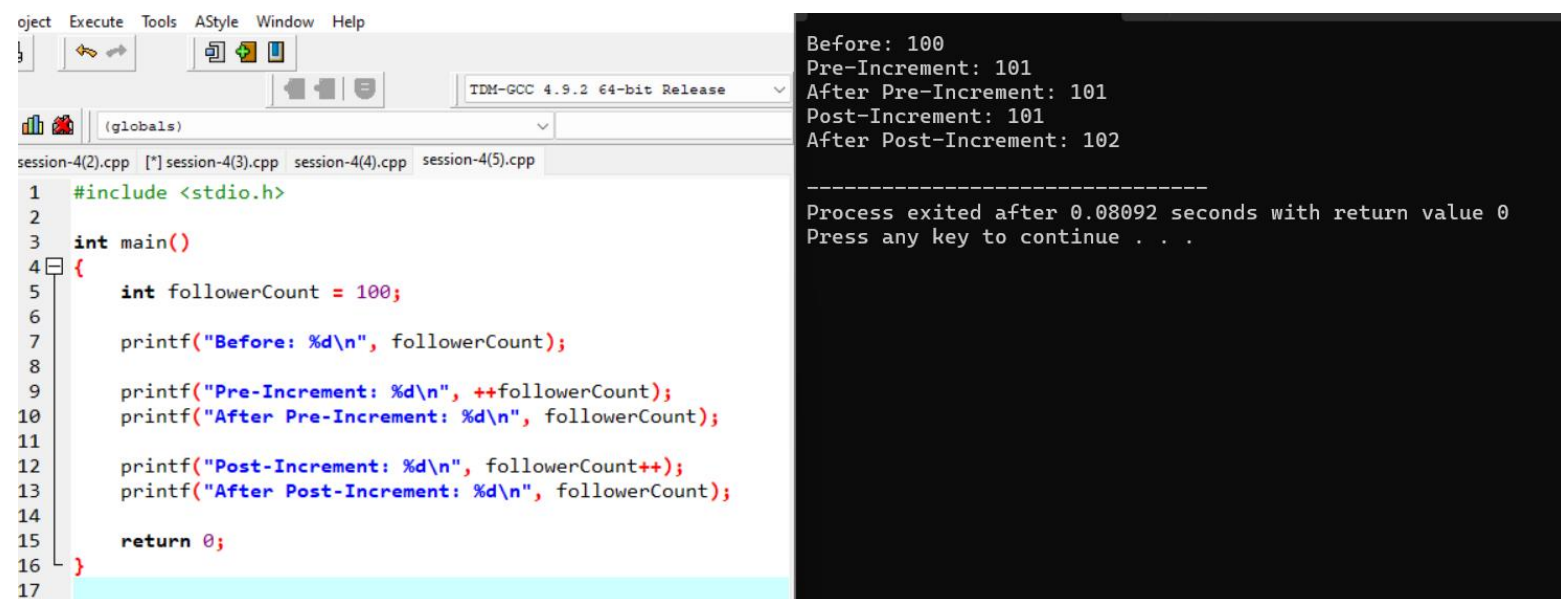
Enter comments: 150

Enter shares: 60

Trending

Q5. Write a code snippet that demonstrates the difference between pre-increment (++count) and post-increment (count++) by logging the values before and after using both on a follower Count variable.

Ans.



The screenshot shows a C++ IDE with a code editor on the left and a terminal on the right. The code in the editor defines a variable `followerCount` and prints its value at various stages of pre-increment and post-increment operations. The terminal output shows the results of these operations.

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int followerCount = 100;
6
7     printf("Before: %d\n", followerCount);
8
9     printf("Pre-Increment: %d\n", ++followerCount);
10    printf("After Pre-Increment: %d\n", followerCount);
11
12    printf("Post-Increment: %d\n", followerCount++);
13    printf("After Post-Increment: %d\n", followerCount);
14
15    return 0;
16 }
17
```

Before: 100  
Pre-Increment: 101  
After Pre-Increment: 101  
Post-Increment: 101  
After Post-Increment: 102

-----  
Process exited after 0.08092 seconds with return value 0  
Press any key to continue . . .

Before: 100

Pre-Increment: 101

After Pre-Increment: 101

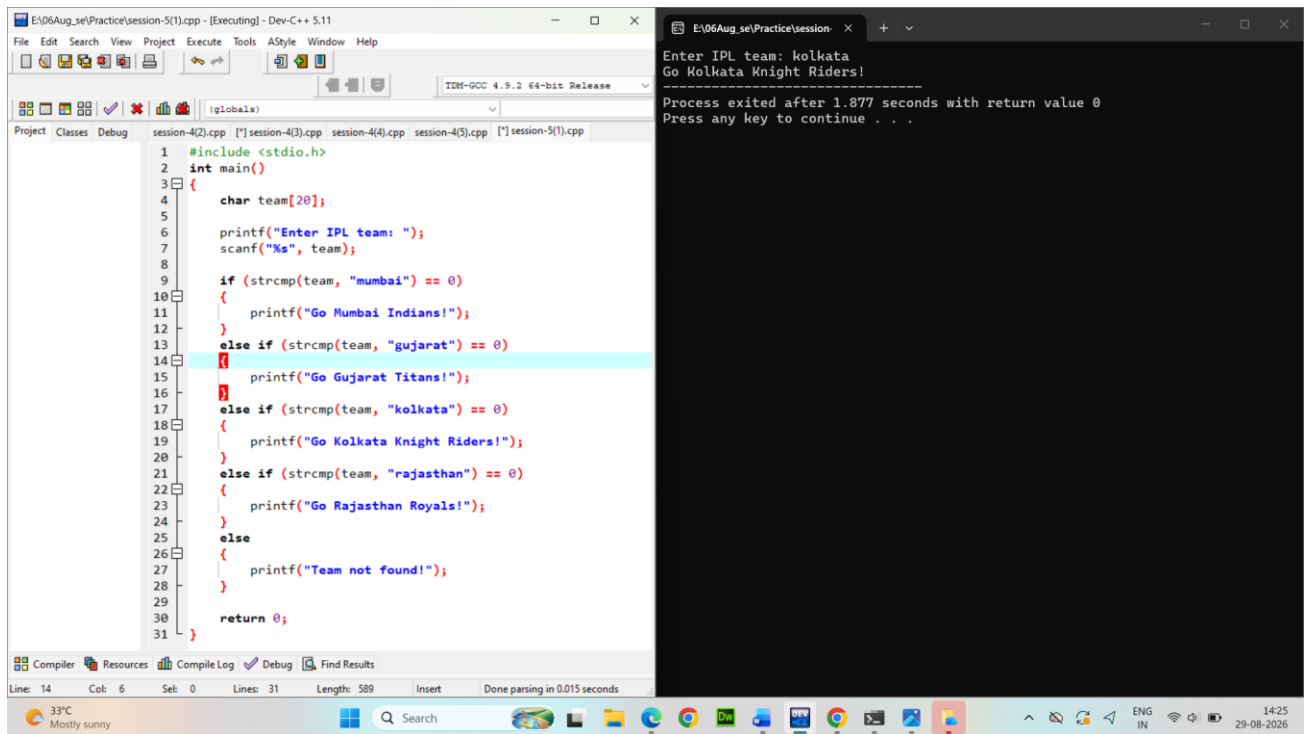
Post-Increment: 101

After Post-Increment: 102

## Session-5

Q1. Create a simple IPL Fan Bot that takes your favorite IPL team name as input and uses if-else-if statements to print a unique cheer message for each team (e.g., 'Go Mumbai Indians!', 'Chennai Super Kings for the win!'). If the team is not recognized, print 'Team not found!'

Ans



The screenshot displays a C++ IDE with the following code in session-5(1).cpp:

```
1 #include <stdio.h>
2 int main()
3 {
4     char team[20];
5
6     printf("Enter IPL team: ");
7     scanf("%s", team);
8
9     if (strcmp(team, "mumbai") == 0)
10    {
11        printf("Go Mumbai Indians!");
12    }
13    else if (strcmp(team, "gujarat") == 0)
14    {
15        printf("Go Gujarat Titans!");
16    }
17    else if (strcmp(team, "kolkata") == 0)
18    {
19        printf("Go Kolkata Knight Riders!");
20    }
21    else if (strcmp(team, "rajasthan") == 0)
22    {
23        printf("Go Rajasthan Royals!");
24    }
25    else
26    {
27        printf("Team not found!");
28    }
29
30    return 0;
31 }
```

The execution output on the right shows:

```
Enter IPL team: kolkata
Go Kolkata Knight Riders!
-----
Process exited after 1.877 seconds with return value 0
Press any key to continue . . .
```

Output:

Enter IPL team: kolkata  
Go Mumbai Indians!

Q2. Build a Zomato-style food suggestion tool: take the user's preferred meal time ('breakfast', 'lunch', 'dinner', or 'snack') and use a switch-case statement to suggest a popular dish for that time. If the input doesn't match any meal, suggest 'Try some fruits!'.

Ans.

```
v Project Execute Tools AStyle Window Help
IDM-GCC 4.9.2 64-bit Release
(globals)
[*] session-5(1).cpp [*] session-5(2).cpp
1  #include <stdio.h>
2  int main()
3  {
4      int choice;
5
6      printf("1. Breakfast\n");
7      printf("2. Lunch\n");
8      printf("3. Dinner\n");
9      printf("4. Snack\n");
10
11     printf("Enter your choice: ");
12     scanf("%d", &choice);
13
14     switch(choice)
15     {
16     case 1:
17         printf("Try Dosa!");
18         break;
19
20     case 2:
21         printf("Try Gujarati Thali!");
22         break;
23
24     case 3:
25         printf("Try Pizza!");
26         break;
27
28     case 4:
29         printf("Try Samosa!");
30         break;
31
32     default:
33         printf("Try some fruits!");
34     }
35 }
```

```
1. Breakfast
2. Lunch
3. Dinner
4. Snack
Enter your choice: snack
Try some fruits!
-----
Process exited after 14.59 seconds with return value 0
Press any key to continue . . .
```

Output

1. Breakfast

2. Lunch

3. Dinner

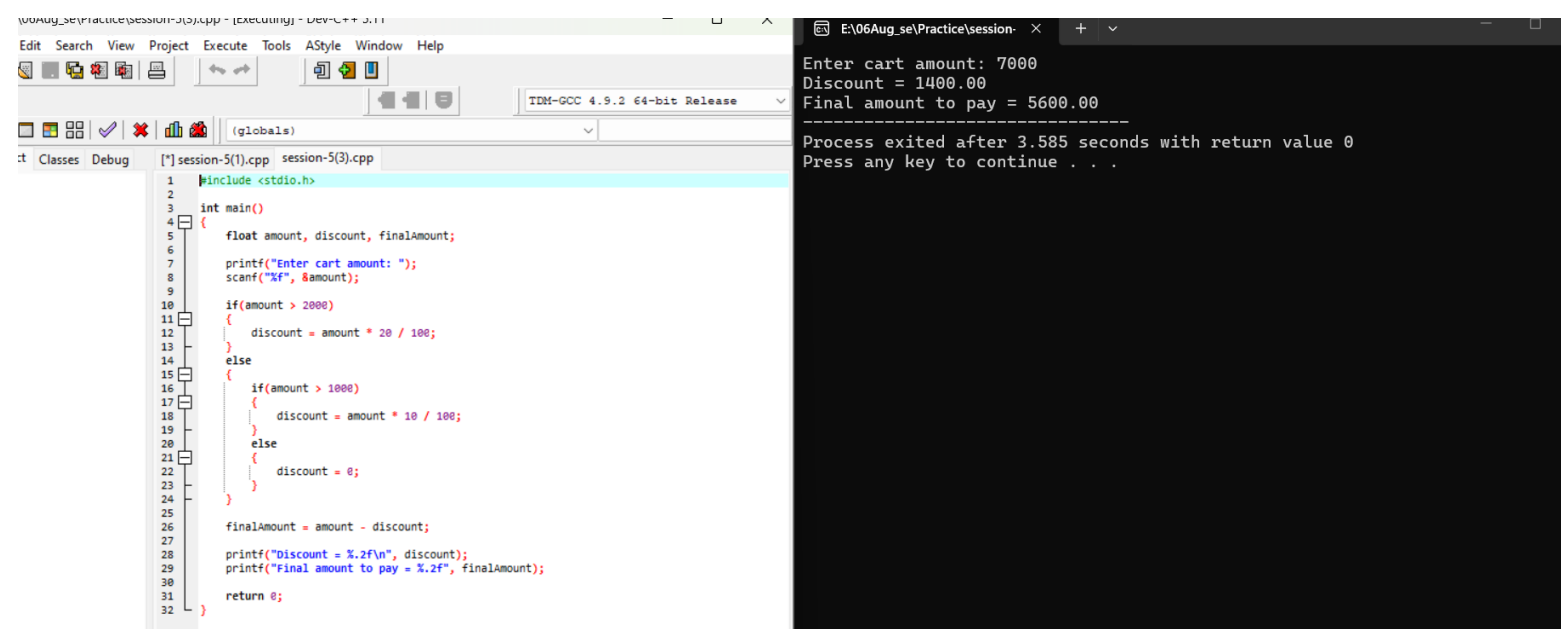
4. Snack

Enter your choice: 2

Try Gujarati Thali!

Q3. Create a Flipkart discount calculator that asks the user for the total cart amount. Use nested if statements to check: if amount > 2000, apply 20% discount; else if amount > 1000, apply 10% discount; else, no discount. Print the final amount to pay.<br><br><em><strong>Hint:</strong> Use nested ifs to check each discount slab.</em>

Ans.



The image shows a C++ IDE with two windows. The left window displays the source code for a discount calculator, and the right window shows the program's output.

```
1 #include <stdio.h>
2
3 int main()
4 {
5     float amount, discount, finalAmount;
6
7     printf("Enter cart amount: ");
8     scanf("%f", &amount);
9
10    if(amount > 2000)
11    {
12        discount = amount * 20 / 100;
13    }
14    else
15    {
16        if(amount > 1000)
17        {
18            discount = amount * 10 / 100;
19        }
20        else
21        {
22            discount = 0;
23        }
24    }
25
26    finalAmount = amount - discount;
27
28    printf("Discount = %.2f\n", discount);
29    printf("Final amount to pay = %.2f", finalAmount);
30
31    return 0;
32 }
```

The output window shows the following text:

```
Enter cart amount: 7000
Discount = 1400.00
Final amount to pay = 5600.00
-----
Process exited after 3.585 seconds with return value 0
Press any key to continue . . .
```

Output

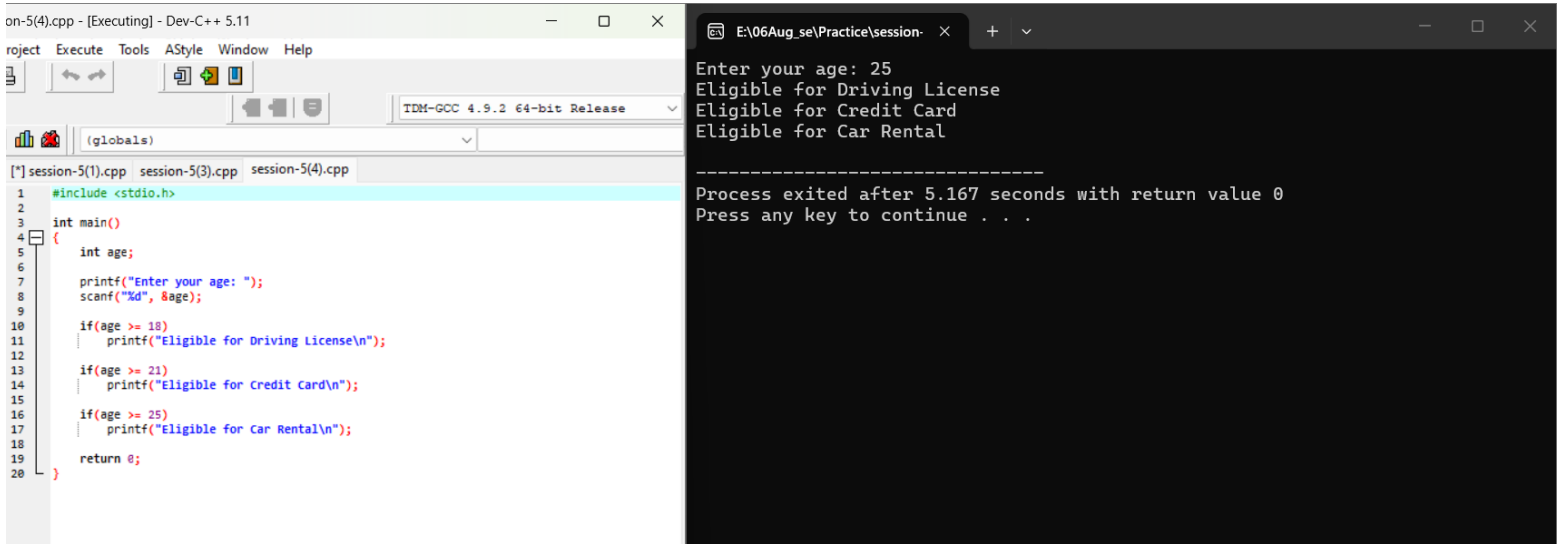
Enter cart amount: 7000

Discount = 1400.00

Final amount to pay = 5600.00

Q4. Write a program that takes a user's age and checks eligibility for three things using if-else statements: if age >= 18, print 'Eligible for Driving License'; if age >= 21, also print 'Eligible for Credit Card'; if age >= 25, also print 'Eligible for Car Rental'. Print all applicable messages for the given age.

Ans



Output

Enter your age: 22

Eligible for Driving License

Eligible for Credit Card

S

